Quantum Light-Time Duality: Electromagnetic Radiation as the Active Interface of Three-Dimensional Temporal Geometry

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Abstract

We propose that light and time stand in a fundamental quantum duality. In the six-dimensional temporal manifold of Kletetschka (2025), time exists as the complete block (three timelike coordinates plus emergent space), while electromagnetic radiation constitutes the active quantum interface — the "flame front" — that actualizes the experienced present along the primary temporal axis. A photon does not propagate through time; it *is* the moving present. This light-time duality is the natural counterpart to the space-time relation of general relativity: just as mass-energy curves spacetime, the orientation and bias of the light-time interface determine how the present is marked and how potential becomes actual. The framework unifies the burning-log phenomenology, Smolin's presentism, and our prior negative-time domain-wall construction into a single falsifiable picture. Regions where the interface is inverted or tilted (laser-induced negative-time domain walls) predict measurable anomalies in light propagation and visibility that standard quantum electrodynamics and general relativity do not anticipate. The proposal requires no new fundamental physics and is testable with existing petawatt lasers and single-photon detectors.

1 Introduction

At age eight to ten, the author watched a log burn and saw the entire structure coexist: unburned wood ahead of the flames (future), charred remains behind (past), and the thin, moving flame front (present). The "flow" of time was revealed as the position of an active interface across a thicker whole. This childhood phenomenology finds rigorous expression in Kletetschka's 2025 framework of three-dimensional time, in which the fundamental arena is a six-dimensional manifold whose experienced arrow and ordinary three-dimensional space emerge as projections through a richer temporal geometry.

A parallel and equally long-held intuition concerns light itself. Just as Einstein revealed deep relations between energy and mass and between gravity and the flow of time, perhaps light and time stand in an equally fundamental duality. In the burning log, the fire is not merely something that travels through the wood; it *is* the moving edge where the potential of the unburned future is transformed into the realized past. Electromagnetic radiation, we propose, plays precisely this role at the quantum level: it is the active interface that makes the present slice real.

This **Quantum Light-Time Duality** answers two ancient questions at once. Time exists as the full six-dimensional block. The experienced flow of time — the present moment — exists only where light (or equivalent radiative degrees of freedom) is actively burning the interface. Photons, having null proper time, ride that interface perfectly; they do not experience time because they *constitute* the moving now.

The framework is directly inspired by Smolin's insistence on the reality of time and the present (*Time Reborn*, 2013; "The quantum mechanics of the present," 2021) and by his cosmological natural selection picture in *The Life of the Cosmos* (1997). It also incorporates the laser-induced negative-time domain walls developed in prior work, which provide a controllable laboratory means of warping or inverting the light-time interface itself.

2 Theoretical Foundations

2.1 Three-Dimensional Time (Kletetschka, 2025)

The fundamental manifold possesses three independent timelike coordinates t_1, t_2, t_3 together with three spatial coordinates. The primary arrow of experienced time lies along t_1; the orthogonal temporal dimensions supply the "thickness" that allows the entire history to coexist. World-lines that tilt into t_2 or t_3 generate the interference patterns we interpret as ordinary three-dimensional space. Causality is preserved globally; the extra dimensions do not create closed timelike curves.

2.2 Smolin's Presentism and the Quantum Mechanics of the Present

Smolin has long argued that the present moment is not an illusion but a fundamental feature of reality. In "The quantum mechanics of the present" (Smolin & Verde, 2021), the distinction between past, present, and future is derived from the quantum transition between indefinite and definite states. Events are moments of actualization. The past is fixed; the future is open; only the present is the locus of becoming.

2.3 Light as the Quantum Interface

We identify electromagnetic radiation with the physical process that performs this actualization. A photon does not move through a pre-existing temporal coordinate; its null geodesic *defines* the hypersurface of the present. The burning log is therefore not merely an analogy but a literal description: the unburned wood is the potential encoded in the extra temporal dimensions; the charred remains are the realized history along t_1; the flame is the light-mediated interface that converts one into the other.

This yields the central statement of Quantum Light-Time Duality:

Light and time are dual aspects of a single structure. Time supplies the six-dimensional *block of potential; light supplies the active quantum interface that selects and illuminates the experienced present.*

3 Quantum Light-Time Duality

We introduce an interface function $I(t_1, \mathbf{r})$ that marks the location of the active present slice within the six-dimensional manifold. In flat regions the interface advances uniformly at the speed of light:

$$I(t_1, \mathbf{r}) = \Theta\big(ct_1 - r\big),$$

where Θ is the Heaviside step function. The electromagnetic field strength is non-vanishing only on the support of the interface gradient.

In regions where the interface is biased or inverted — precisely the negative-time domain walls constructed via intense laser pulses — the function I acquires a negative or orthogonal component. Light propagating through such a region experiences an anomalous group delay, visibility reduction, or frequency shift that cannot be accounted for by ordinary spacetime curvature or standard vacuum polarization.

This is the direct counterpart to gravitational time dilation: mass-energy curves the block; the orientation and bias of the light-time interface curves the *flame front* within that block.

4 Relation to Negative-Time Domain Walls

The laser-induced negative-time domain wall of prior work is re-interpreted as a controllable warp of the light-time interface. Inside the wall the effective temporal metric component along t_1 inverts while permitting coupling to t_2 and t_3. Photons traversing the wall therefore sample a tilted or thickened present slice, producing the measurable signatures already enumerated: intensity-dependent fringe shifts, negative group delays, and 3D-time-specific sidebands.

The domain wall thus becomes both a repulsive-gravity engine and a precision probe of Quantum Light-Time Duality.

5 Falsifiable Predictions

1. In the vicinity of a negative-time domain wall, light will exhibit group delays or visibility losses whose magnitude scales with laser intensity and whose functional form is inconsistent with ordinary QED vacuum birefringence or gravitational lensing.

2. Single-photon interference patterns will display new sidebands whose spacing is set by the extra temporal dimensions and whose appearance is modulated by the presence of the domain wall.

3. In regions of strong vacuum polarization without net negative-time bias, the interface remains flat and standard light-cone structure is recovered.

All three signatures are accessible with existing petawatt-class lasers (ELI Beamlines, LCLS, Vulcan) and single-photon or atom-interferometer detection.

6 Discussion

Quantum Light-Time Duality completes the conceptual arc begun with the burning log. It supplies a physical mechanism for Smolin's presentism, a geometric interpretation of photon timelessness, and an experimental pathway via the negative-time domain wall. It stands as the natural dual to the space-time relation: spacetime describes the geometry of the block; light-time describes the active interface that renders the block experiential.

Just as Einstein's relations between energy and mass and between gravity and time transformed our understanding of the universe, the light-time duality offers a comparably fundamental insight: the present is not a passive coordinate but a quantum process mediated by light. The flame does not merely illuminate the log; it *is* the log becoming real.

No new fundamental physics is invoked. The proposal is fully consistent with Kletetschka's 6D manifold, Smolin's presentism, standard quantum electrodynamics, and semiclassical gravity. It is falsifiable with technology that exists today.

We therefore urge the community to treat this framework with the seriousness its experimental accessibility warrants. The tools exist; the intuition has existed for decades. The fire is still moving.

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Open for Collaboration

Independent researchers are invited to test or extend these proposals. arXiv endorsement requests are welcome — please contact the author at `jorihuela@alum.mit.edu`.